

3(a)	1 <i>correct axis labels:</i> percentage / %, cover of, (water) hyacinth and percentage / %, cover of, malancha ; 2 appropriately shaped line ; straight line with upward trend	2
3(b)(i)	any suitable method of generating random numbers and matching to the (69) sampling locations ;	1
3(b)(ii)	48 (%) ; ; award one mark for correct working if final answer incorrect or missing	2
3(b)(iii)	<i>t</i>-test and <i>any one from:</i> <i>idea of</i> comparing (the difference between two) means ; continuous data ; standard deviations (for each treatment) are approximately the same ; data (from populations that are) normally distributed ; fewer than 30 values (in each treatment) ;	1

Question	Answer	Marks																					
3(b)(iv)	ref. to hazard and risk and precaution ; <table> <tr> <th>hazard</th><th>risk</th><th>precaution</th></tr> <tr> <td>animals</td><td>allergy / bites / injury / infection / disease</td><td>medication / work in a group / travel with an expert / suitable PPE / first aid kit</td></tr> <tr> <td>(plants) pollen</td><td>allergy</td><td>medication / mask</td></tr> <tr> <td>environment woodland / falling branches or trees / thorns / trip hazards / holes / mud</td><td>injury / getting lost</td><td>boots / hard hat / suitable clothing / PPE / GPS / map / stay in a group / use an expert /</td></tr> <tr> <td>water / river / Armand bayou</td><td>injury / drowning / hypothermia / swept away (by current)</td><td>boots / lifejacket / wetsuit / use an expert / do not enter water / use a boat / use a rope</td></tr> <tr> <td>(water-borne) pathogens / parasites / pollution</td><td>infection / disease / irritation (from pollutants)</td><td>PPE / wash hands</td></tr> <tr> <td>tools / knife (to cut plants)</td><td>injury</td><td>cut away from, hand / body / other people</td></tr> </table>	hazard	risk	precaution	animals	allergy / bites / injury / infection / disease	medication / work in a group / travel with an expert / suitable PPE / first aid kit	(plants) pollen	allergy	medication / mask	environment woodland / falling branches or trees / thorns / trip hazards / holes / mud	injury / getting lost	boots / hard hat / suitable clothing / PPE / GPS / map / stay in a group / use an expert /	water / river / Armand bayou	injury / drowning / hypothermia / swept away (by current)	boots / lifejacket / wetsuit / use an expert / do not enter water / use a boat / use a rope	(water-borne) pathogens / parasites / pollution	infection / disease / irritation (from pollutants)	PPE / wash hands	tools / knife (to cut plants)	injury	cut away from, hand / body / other people	1
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3(b)(v)	any four from: <i>in context of supports:</i> 1 (removal of malancha) reduces, (mean) <u>area / abundance</u> (of water hyacinth) in treatment 1 and 2 (compared to, control / treatment 3) or (mean) <u>area / abundance</u> (of water hyacinth) is, lower / lowest / AW, in treatment 2, compared to, control / treatment 3 ; 2 (95%) confidence intervals / CI / error bars, for treatment 2 and 3, do not overlap and (mean area / abundance of water hyacinth) is significantly different ; <i>in context of limits support:</i> 3 (95%) confidence intervals / CI / error bars, for treatment 1 and 3, overlap and (mean area / abundance of water hyacinth) may not be significantly different ; 4 investigation only carried out, once / at one time of year / over period of 8 weeks ; 5 investigation only carried out at 12 locations ; 6 no statistical test / should carry out a statistical test ;	4